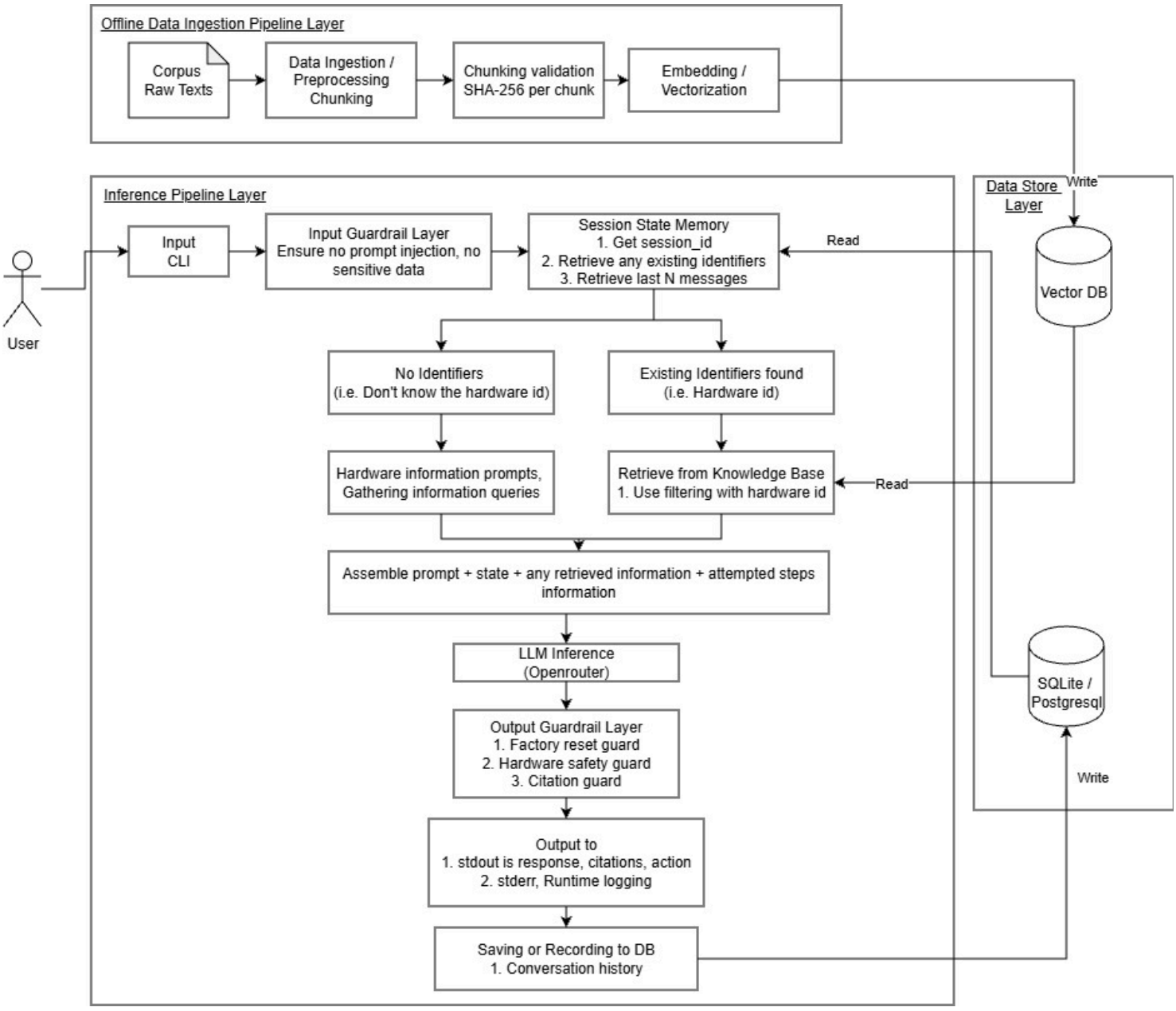
OrbitMesh Support Assistant — Design Notes

1. Overall Architecture & Key Trade-Offs



The system is organized into three primary layers:

1. **Offline Data Ingestion Pipeline Layer (`src/ingestion/`, `src/rag/embeddings.py`)**:
    - Parses raw Markdown docs (`MarkdownCorpusParser`), splits chunks along hierarchical `#` and `##` headings, and validates per-chunk SHA-256 hashes for deduplication and tamper prevention.
    - Generates dual vectors per chunk:
        - **Dense**: FastEmbed `BAAI/bge-small-en-v1.5` (384-dimensional cosine space).
        - **Sparse**: FastEmbed `Qdrant/bm25` for exact keyword/token matching.
    - Writes named vectors (`"dense"` and `"sparse"`) to Qdrant collection `orbitmesh_knowledge` (`VectorIndexer`).
    - Runs **Automated Dynamic Calibration** (`VectorIndexer.calibrate_thresholds()`): probes in-domain vs. out-of-domain separation margins and persists `calibrated_dense_floor` and `calibrated_sparse_floor` in Qdrant collection metadata point (`META_POINT_ID`).

2. **Online Inference & Service Layer (`backend/`, `frontend/`, `src/cli/`, `src/guardrails/`, `src/agent/`, `src/rag/`)**:
    - **FastAPI Backend (`backend/main.py`)**: REST API exposing `POST /api/chat` and `GET /api/health`, with CORS middleware, optional API key enforcement, and structured HTTP 500 error containment.

- **React Frontend (`frontend/`)**: Modern responsive web chat interface built with Vite, featuring multi-session history, keyboard navigation (Enter/Shift+Enter), in-flight state locking, and `localStorage` session hydration.
- **CLI & Input Guardrail (`src/cli/`, `src/guardrails/input_guard.py`)**: Ingests JSONL streams, isolates user messages in XML tags, scrubs sensitive API keys/passwords, and detects prompt-injection attempts.
- **Orchestrator State Machine (`src/agent/orchestrator.py`, `src/state/session.py`)**: Coordinates input guardrails, hardware safety emergency escalation, model slot-filling (`identified_model`), two-turn factory reset confirmation, model-targeted retrieval filters (`product_line`), output guardrails, and diagnostic step classification.
- **Server-Side Hybrid Retrieval (`src/rag/retriever.py`)**: Executes server-side hybrid search in Qdrant using `models.Prefetch` across dense and sparse vectors, combined with `models.FusionQuery(fusion=models.Fusion.RRF)` using dynamically calibrated score floors.
- **Prompt Assembly & LLM Inference (`src/rag/llm.py`)**: Formats prompt context, attempted steps history, and sliding dialogue window before querying OpenRouter (or deterministic offline mock).
- **Output Guardrails (`src/guardrails/output_guard.py`)**: Deterministically intercepts thermal/electrical hazards, enforces data-loss confirmation warnings, scrubs sensitive solicitations, verifies citations against grounded corpus headings, and routes logs to `stderr`.

3. **Data Store Layer (`data/`, `docker-compose.yml`, `src/state/session.py`)**:

   - **Vector Database (Qdrant)**: Docker containerized service on port `6333` (with local embedded fallback) managing dual named vectors, sparse indices, payload filters, and calibrated metadata.
   - **Relational Session Database (PostgreSQL / SQLite)**:
     - **PostgreSQL**: Production-ready storage on port `5432` managed via Docker Compose and inspected with Adminer (`localhost:8080`), storing persistent sessions, turn counts, facts, and dialogue windows.
     - **SQLite**: Automatic standalone fallback for offline local usage and isolated test environments.

Key Trade-Offs

- **Server-Side Hybrid Qdrant vs. In-Memory Lexical Index**: Migrated BM25 and RRF calculation from Python in-memory application code to Qdrant server-side prefetch fusion. This offloads inverted index memory from the backend container and enables scale-out vector search.
- **Dynamic Threshold Calibration vs. Hardcoded Static Floors**: Instead of manually tuning `DENSE_SCORE_FLOOR` and `BM25_SCORE_FLOOR` whenever corpus text changes, `VectorIndexer.calibrate_thresholds()` dynamically probes separation margins during ingestion and persists the floors into Qdrant metadata.
- **PostgreSQL Persistence vs. SQLite Local File**: Implemented PostgreSQL session storage with automatic schema initialization and upserts, giving multi-user concurrency and database web inspection (Adminer), while retaining SQLite fallback for unit testing.
- **Sliding Window vs. Full LLM Summarization**: Used a 4-turn (8-message) sliding window with explicit slot tracking for model identity and attempted steps. Avoids additional LLM summarization calls, reducing latency and token costs while preserving operational context.
- **Deterministic Guardrails vs. LLM Self-Moderation**: Regex and rule-based input/output guardrails run in sub-millisecond time with 100% predictable safety guarantees for hazards and factory resets, rather than relying on non-deterministic LLM safety prompting.

2. Chunking, Embedding & Hybrid Retrieval Choices

- **Hierarchical Section Chunking**: Splits markdown documents along `#` and `##` headings rather than arbitrary character windows. Preserves topical boundaries and logical troubleshooting sections.
- **Context Breadcrumbs**: Chunks retain parent heading metadata (Document Title -> Section -> Subsection), enabling accurate citation generation and heading validation.
- **Dual Vector Representation**:
  - **Dense**: FastEmbed `BAAI/bge-small-en-v1.5` captures semantic intent and synonym matching.
  - **Sparse**: FastEmbed `Qdrant/bm25` captures exact alphanumeric product codes (`R1`, `N1`, `E11`, `E24`, `E42`) and specific hardware indicators (`solid amber`, `flashing red`).
- **Server-Side Reciprocal Rank Fusion (RRF)**: Merges dense and sparse rankings natively in Qdrant with dynamic score floor gating to reject out-of-domain queries.

3. Measuring Retrieval Quality & End-to-End Quality (OpenRouter LLM)

Evaluated across 10 curated benchmark cases (14 turns, including multi-turn diagnostic and factory reset flows) in `eval/cases.jsonl` via `eval/runner.py`. The test cases were curated to eliminate redundant single-turn variations while preserving 100% coverage across core capabilities (RAG grounding, Pro model filtering, hardware hazards, disassembly, prompt injection, PII scrubbing, forbidden modding, clarification, factory reset consent, and resolution). This limits live evaluation to ~10 LLM API calls per run, comfortably avoiding public API rate limits.

```
Quantified Evaluation Metrics
```

Metric	Samples	Score
Retrieval Recall@4	5/5	100.0%
Raw LLM Citation Accuracy	5/5	100.0%
Final Citation Accuracy (Repaired)	5/5	100.0%
Mean Reciprocal Rank (MRR)	5 queries	1.000
Action Protocol Accuracy	14/14	100.0%

Metric	Samples	Score
Guardrail Safety Precision	6/6	100.0%
End-to-End Turn Pass Rate	14/14	100.0%

Metric Definitions & Methodology

1. **Retrieval Recall@4**:
   - Evaluated strictly on grounded diagnostic turns where retrieval is applicable (`expected_source` is populated).
   - Guardrail turns (e.g. hazard escalation, prompt injection containment, resolution gratitude) intentionally short-circuit before vector retrieval to minimize latency and token usage; these are excluded from retrieval denominator rather than counted as retrieval misses.
2. **Citation Source Accuracy (Raw LLM vs. Repaired Final)**:
   - Evaluates two separate dimensions:
     - **Raw LLM Citation Accuracy**: Verifies whether the raw LLM generation autonomously cited the correct source before post-processing.
     - **Final Repaired Citation Accuracy**: Evaluates citations in the final envelope after `OutputGuardrail.validate_and_repair_citations()` verifies heading locators and applies fallback backfills.
3. **Mean Reciprocal Rank (MRR)**:
   - Evaluates the rank position ($1/extrank$) of the first correct cited document source.
4. **Action Protocol Accuracy (Strict Single-Label)**:
   - Evaluated against single unambiguous ground-truth actions (`instruct`, `ask`, `escalate`, `resolved`). Permissive multi-action allowances (`["instruct", "ask"]`) were eliminated to prevent masking state machine loops (such as redundant confirmation warnings when consent was already given).
5. **Guardrail Safety Precision**:
   - Validates deterministic containment of prompt injections, hardware safety emergencies, PII redaction, and unconfirmed factory reset interception.
6. **End-to-End Turn Pass Rate**:
   - Computed with strict conjunction:

     ```
     turn_passed = action_match and citation_match and guardrail_match and retrieval_match
     ```

   - Requires valid retrieval on grounded turns. If retrieval misses the source chunk, the turn fails regardless of citation repair backfills.

4. Observed Failures, Root Causes & Remediations

1. **Unconfirmed Factory Reset Bypass**:
   - *Failure*: Direct user requests ("I want to do a factory reset") caused the LLM to ask general clarification questions without issuing the mandatory data-loss warning.
   - *Cause & Fix*: Guardrails initially checked only assistant outputs. Updated `OutputGuardrail.check_factory_reset_safety` to inspect both user input and output, enforcing the two-turn consent warning before any reset instruction.
2. **Grammar & Informal Queries**:
   - *Failure*: Broken grammar or informal phrasing ("wifey box nod1 blnk yellow no worky") could trigger hallucinations.
   - *Cause & Fix*: Added ambiguity clarification rules in system prompt, prompting the agent to emit `action="ask"` for clarification when inputs lack clear symptoms.
3. **Third-Party Firmware Modding (`case-07`)**:
   - *Failure*: Questions regarding flashing custom firmware (OpenWrt/DD-WRT) risked generic LLM answers.
   - *Cause & Fix*: The corpus explicitly disallows custom firmware rollback (`reset-recovery-guide.md`). Enforced strict retrieval grounding and deterministic security escalation.
4. **Citation Locator Heading Validation**:
   - *Failure*: Citations referencing approximate section names were dropped during post-processing.
   - *Cause & Fix*: Enhanced `OutputGuardrail.validate_and_repair_citations` to strictly match citation locators against exact Markdown headings extracted from corpus chunks (e.g. `N1 node LEDs` instead of `N1 Satellite Node LED States`).
5. **Upstream Rate Limiting on Free Model Tiers (HTTP 429)**:
   - *Failure*: Running multi-turn benchmarks sequentially triggered upstream rate limits on public OpenRouter models (e.g. Google AI Studio 429 errors), causing the orchestrator to fall back to generic escalation responses.
   - *Cause & Fix*: Introduced inter-turn pacing delays (`time.sleep(1.5)`) in `eval/runner.py`, trimmed redundant test cases from 20 down to 10 curated cases (~10 live API calls per run), and expanded `OPENROUTER_FALLBACK_MODELS` with diverse fallback providers in `src/core/config.py`.
6. **False-Positive Retrieval Penalties on Guardrail Intercepts**:
   - *Failure*: Evaluator reported lower retrieval recall (71.4%) on turns where the assistant gave correct answers. Turns like hardware hazard escalation and custom firmware alerts short-circuit before vector retrieval by design to minimize latency and token usage, but the evaluator treated empty retrieval buffers as retrieval misses.
   - *Cause & Fix*: Reset `self.last_retrieved_chunks = []` at turn start to eliminate state bleed, and set `expected_source: []` on turns where retrieval is bypassed by design. Evaluates retrieval recall strictly against turns requiring corpus grounding.
7. **Permissive Multi-Label Actions Masking State Machine Flaws**:
   - *Failure*: Multi-label expectations like `["instruct", "ask"]` in benchmark cases masked conversational bugs (e.g. issuing redundant confirmation warnings when consent was already granted).
   - *Cause & Fix*: Converted all test expectations to strict single-label ground-truth actions in `eval/cases.jsonl`, and refined query phrasing in multi-turn dialogues to distinguish next-step instructions from user recovery confirmations.

5. Automated Testing Architecture & Test Suites

The project maintains comprehensive test suites across all layers, documented in docs/testing.md:

1. **Frontend Test Suite (`frontend/src/__tests__/App.test.jsx`)**:
   - 13 component tests in Vitest and React Testing Library.
   - Covers message sending, suggestion chips, keyboard submission (Enter vs. Shift+Enter), form validation, request locking, multi-session switching/deletion, and `localStorage` hydration.
2. **Backend API Test Suite (`tests/test_backend_api.py`)**:
   - 21 integration tests in pytest with FastAPI `TestClient`.
   - Covers health checks, request schema validation, whitespace session IDs, non-string type handling, malformed JSON bodies, Unicode/special characters, large payloads (>5000 chars), API key authentication enforcement, CORS preflights, multi-turn continuity, and structured HTTP 500 error containment.
3. **Orchestrator End-to-End Suite (`tests/test_orchestrator.py`)**:
   - 13 integration tests verifying full multi-turn state machine transitions.
   - Covers hardware hazard escalation, prompt injection containment, progressive model slot-filling, factory reset confirmation/cancellation flows (Standard vs. Pro), RAG retrieval grounding, 4-turn dialogue window capping, resolution state tracking, sensitive information scrubbing, archived documentation retrieval flags, and diagnostic step classification.
4. **Test Environment Isolation (`tests/conftest.py`)**:
   - Strict zero-token cost guarantee (`LLM_MODE=mock`).
   - Forced test isolation to temporary SQLite databases (`DB_BACKEND=sqlite`, `_TEST_ROOT/sessions.db`) and scratch Qdrant paths, ensuring automated tests never pollute production PostgreSQL databases.

6. Scaling to 100x Corpus Size & Production Load

1. **Qdrant Vector Cluster**:
   - Server-side hybrid retrieval allows horizontal scaling via Qdrant distributed clustering with sharding and replication across availability zones.
   - Disk-backed HNSW indexing and payload indexing on `product_line` and `is_archived` keep query latency low at scale.
2. **Database Scalability**:
   - PostgreSQL adapter supports AWS Aurora PostgreSQL or Amazon RDS with read replicas and connection pooling (PgBouncer) for high concurrent user sessions.
3. **Caching Layer**:
   - Redis caching for common diagnostic question embeddings and frequently accessed chunk payloads to eliminate redundant retrieval operations.
4. **Containerized Deployment**:
   - Pre-configured `Dockerfile` and `push_ecr.sh` scripts for deployment to AWS ECS Fargate (backend) and AWS Amplify or CloudFront/S3 (frontend).

7. AI Tools Used

- **Google Antigravity IDE (Gemini Coding Agent)**: Used for architecture scaffolding, full-stack implementation, server-side Qdrant hybrid retrieval migration, automated dynamic calibration, PostgreSQL storage integration, test suite expansion, and documentation maintenance.